

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

PRINCIPLES OF ELECTRONIC COMMUNICATION

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

Module Outcome Cognitive level

1	Define Amplitude modulation	M1.01	R
2	The bandwidth of AM modulated wave is the highest frequency of message signal.	M1.04	R
3	Write the expression for AM modulated wave	M1.03	R
4	The equation for calculating sampling rate is	M2.01	R
5	Define PCM	M2.02	R
6	Name the circuit used in transmitter for boosting the high frequency signals.	M3.01	R
7	Define Signal to Noise ratio	M3.03	R
8	Define selectivity in radio receiver	M4.01	R
9	The value of IF in AM receiver is	M4.02	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

Module Outcome Cognitive level

1	Name different types of modulation techniques.	M1.01	R
2	Compare different parameters of DSBSC and SSB systems in AM.	M1.02	U
3	Compare various pulse modulation techniques.	M2.02	U
4	State sampling theorem and write its mathematical expression.	M2.01	R
5	List the applications of microwave antenna.	M2.03	R
6	Draw the block diagram of high-level AM transmitter	M3.01	R
7	List the advantages of Collector modulation.	M3.02	R
8	Explain the working of balanced modulator.	M3.02	U

9	Explain the choice of IF in radio receivers.	M4.02	U
10	List the applications of frequency modulation.	M4.02	R

PART C

III. Answer all questions from the following

(6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Derive the expression for AM modulated signals and draw the AM wave.	M1.02	U
OR			
2	Compare AM and FM	M1.01	U
3	Derive the expression for 1. The total power transmitted in AM. 2. Modulation index in terms of $V_{\max}$ and $V_{\min}$	M1.02	U
OR			
4	Derive the expression for FM modulated signals and draw the FM wave.	M1.03	U
5	Describe the generation of PWM signals	M2.02	U
OR			
6	Describe Microwave antennas and its applications.	M2.03	U
7	Describe the block diagram of AM Transmitter	M3.01	U
OR			
8	Explain internal noises generated in communication systems	M3.03	U
9	Explain the working of FM superheterodyne radio receiver	M4.02	U
OR			
10	Describe the working of AM diode detector circuit	M4.03	U
11	Explain the working of AM superheterodyne radio receiver	M4.02	U
OR			
12	List and explain any three characteristics of radio receiver.	M4.01	U